

AI4Mobility

A multimodal assistive platform for parents supporting children with mobility difficulties. This deck covers the classifier: three model families, how they were chosen, and how they were judged.

PRESENTED BY

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What this deck covers

Models, protocol, results, and the parts of the result that deserve caution.

1

The data I was given

Synthetic, subject-split, 45 features — and what that permits me to claim.

2

Three model families

Why three rather than one, and the hyperparameter choices behind each.

3

The evaluation protocol

Five-fold cross-validation on training data only, and a test set scored once.

4

Results in detail

Per-class precision, recall and F1; the generalisation gap; confusions; feature importance.

5

Limitations and next steps

What the comparison cannot settle, and what would settle it.

Team Deepminds — four roles, one platform

Four members, four defined responsibilities. This deck presents the modelling and evaluation work.

Krushna Sai Teja Adhala

Data engineering and pipeline lead

Peer contribution split: 28%

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THIS DECK

Model development and evaluation lead

Peer contribution split: 26%

Nandi Reddy Shashidhar Reddy

Pose estimation and visualisation lead

Peer contribution split: 24%

Dhanush Sanjay

Research and documentation lead

Peer contribution split: 22%

The dataset is synthetic

Every accuracy figure in this deck is a property of a seeded generator, not evidence about children.

What the results license us to claim

- The three model families were compared fairly, on identical data and identical splits.
- The evaluation protocol is sound and would transfer unchanged to a recorded dataset.
- The differences between models are real differences on this data.
- The error structure matches the structure deliberately built into the generator.

What they do not license

- No claim that 0.928 accuracy would hold on real children.
- No clinical claim, and no claim of diagnostic value.
- No comparison against published results on recorded datasets.
- No claim that the model is ready to give feedback unsupervised.

01

Three model families

Bagged trees, boosted trees and a neural network — chosen so the comparison is not confined to one kind of model.

Why three models, not one

1

A single model cannot tell you what is data and what is method

When all three families agree on which classes are weak, the weakness belongs to the data rather than to an algorithm.

2

Random Forest: bagged trees, the project baseline

Handles mixed feature scales, exposes feature importance, and is straightforward to constrain.

3

Gradient Boosting: a stronger tree-based comparison

Tests whether a more aggressive tree method buys anything on this feature table.

4

Neural Network (MLP): a different family entirely

Included so the comparison is not confined to trees, and so a non-tree view of the same features is available.

THE FINDING THIS DESIGN PRODUCED

All three models name `lower_limb` and `thread_the_needle` as their weakest classes. That agreement is the evidence that the limit is in the data, not in the choice of algorithm.

Every choice has a stated reason

Settings were fixed before the test set was touched, and none was tuned against it.

Model	Settings	Why
Random Forest	n_estimators=300, max_depth=12, min_samples_leaf=2, class_weight="balanced"	Depth constrained on purpose; balanced weights because the classes are unevenly represented
Gradient Boosting	n_estimators=150, learning_rate=0.1, max_depth=3	Shallow trees with a moderate rate — the standard defensive setting for a boosted ensemble
Neural Network (MLP)	hidden_layer_sizes=(128, 64), alpha=1e-3, max_iter=1500	Two hidden layers for 45 features; iterations raised until convergence warnings stopped

All three models share random_state=42, so every result in the report reproduces exactly.

ONE SETTING THAT NEEDED EXPLAINING

Early stopping is left off for the MLP, because this scikit-learn version's internal validation scoring fails on string class labels. Convergence is monitored through the loss curve instead.

Random Forest max_depth was constrained to 12

The unconstrained forest scores better on training data and worse where it matters.

What unconstrained depth does

- Training accuracy rises to approximately 1.000.
- Generalisation gets noticeably worse.
- Nothing in the output signals that anything went wrong.
- The failure is invisible unless train and test accuracy are reported together.

What the constraint bought

- Training accuracy 0.965 rather than near-perfect memorisation.
- Test accuracy 0.928 and macro F1 0.935.
- A generalisation gap of +0.037, against +0.077 for both other models.
- A model whose training score is a believable description of what it learned.

THE PRINCIPLE

A perfect training score is a warning, not an achievement. Constraining depth costs a little training accuracy and buys a model that behaves the same way on data it has not seen.

02

The evaluation protocol

Cross-validation on training data only, a test set scored once, and per-class metrics reported alongside accuracy.

Five-fold cross-validation, then one test score

- 1 Cross-validation runs on the training partition only**
Five folds, scored on macro F1, used for every model selection decision.
- 2 The test set was scored exactly once**
After all model selection was complete. It was never used to choose a setting.
- 3 The split is by subject, not by frame**
577 training frames and 195 test frames, with whole subjects on one side or the other.
- 4 The scaler is fitted on training data only**
Fitting before splitting would leak test statistics into training, and nothing in the metrics would show it.
- 5 Identical inputs for all three models**
Same features, same splits, same seed, so differences are attributable to the model.

WHY THIS IS STATED SO PROMINENTLY

Cross-validated macro F1 of 0.955 for the Random Forest sits above its test score of 0.935. Reporting both is what makes the difference between the two visible rather than hidden.

Per-class metrics, not headline accuracy

One number for a six-class problem hides the failure that would matter most to a parent.

What accuracy alone hides

- A class that is quietly wrong is worse than a slightly lower overall score.
- The largest class can carry the headline while a small class fails.
- calf_control has a test support of only 14 frames — its perfect score needs that caveat attached.
- Accuracy says nothing about which mistakes are being made.

What we report instead

- Precision, recall and F1 for every class, with support.
- Macro F1 alongside accuracy, so small classes count equally.
- Train accuracy and test accuracy together, so the gap is visible.
- The full confusion matrix, so the shape of the errors is inspectable.

03

Results

Three models within half a point of each other on accuracy, and a clear difference in how well they generalise.

Random Forest wins narrowly, and generalises best

Three model families, identical data and identical splits. Measured on synthetic data.

Model	Test accuracy	Test macro F1	CV macro F1 (5-fold, train only)	Train accuracy	Generalisation gap
Random Forest	0.928	0.935	0.955 ± 0.017	0.965	+0.037
Gradient Boosting	0.923	0.931	0.939 ± 0.016	1.000	+0.077
Neural Network (MLP)	0.923	0.932	0.939 ± 0.019	1.000	+0.077

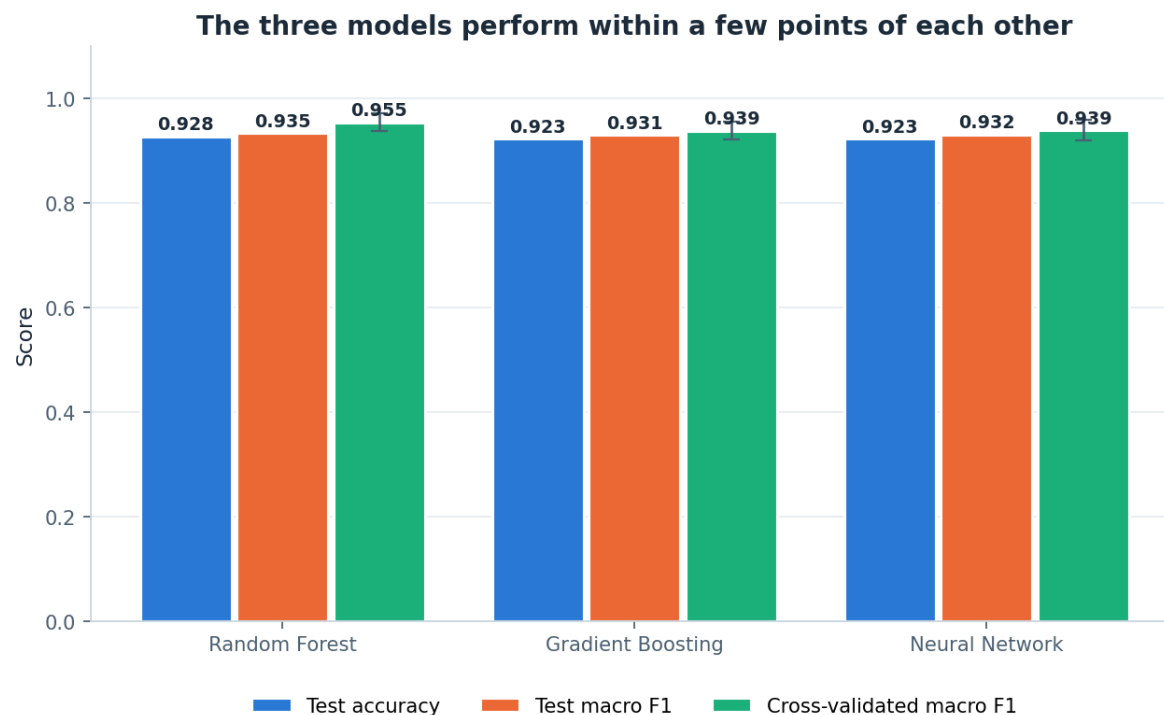
Cross-validation is five-fold on the training partition only. The test set was scored once, after all model selection was complete.

HOW THE WINNER WAS CHOSEN

Random Forest leads on macro F1 at 0.935 by a small margin, and on the generalisation gap by a large one: +0.037 against +0.077 for both alternatives.

FIGURE 6

The three models perform within a few points of each other



Cross-validated scores are five-fold on the training partition only; the error bar is one standard deviation. The test set was scored once, after all model selection was complete.

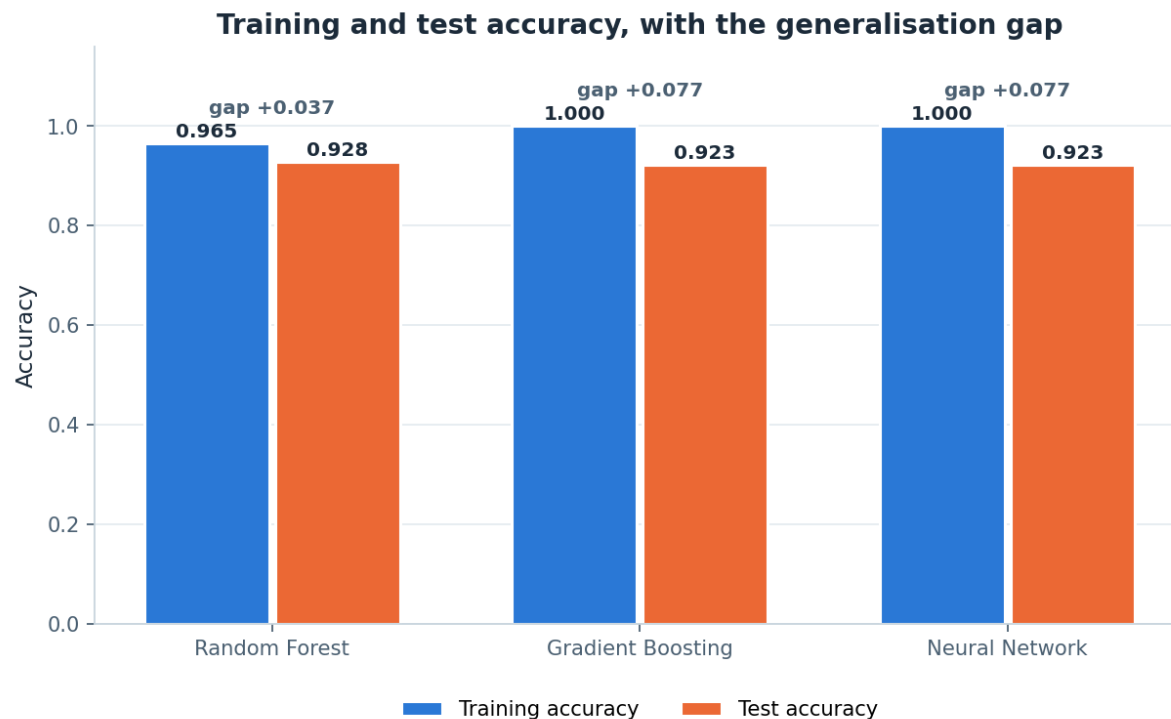
What the bars show

- Test accuracy differs by half a percentage point across the three families.
- Cross-validated macro F1 is higher than the test score for every model.
- The error bar is one standard deviation across the five folds.
- On a comparison this close, the tie-breaker has to be something other than accuracy.

Figure 6 — test accuracy, test macro F1 and cross-validated macro F1 for the three models.

FIGURE 7

The generalisation gap is the honest tie-breaker



Random Forest depth was constrained deliberately. Left unrestricted it fits the training data almost perfectly and generalises worse, with nothing in the output to signal that anything went wrong.

Figure 7 — train against test accuracy, with the gap annotated for each model.

Reading the gap

- Gradient Boosting and the MLP both reach 1.000 training accuracy.
- Both carry a +0.077 gap between training and test.
- The constrained Random Forest trains to 0.965 and carries +0.037.
- A model that memorises its training data will fail quietly on a new subject.

WHY THIS IS VISIBLE AT ALL

The subject-level split is what makes the gap meaningful. On a frame-level split all three gaps would have looked smaller and none of them would have meant anything.

FIGURE 9

Per-class F1 for all three models

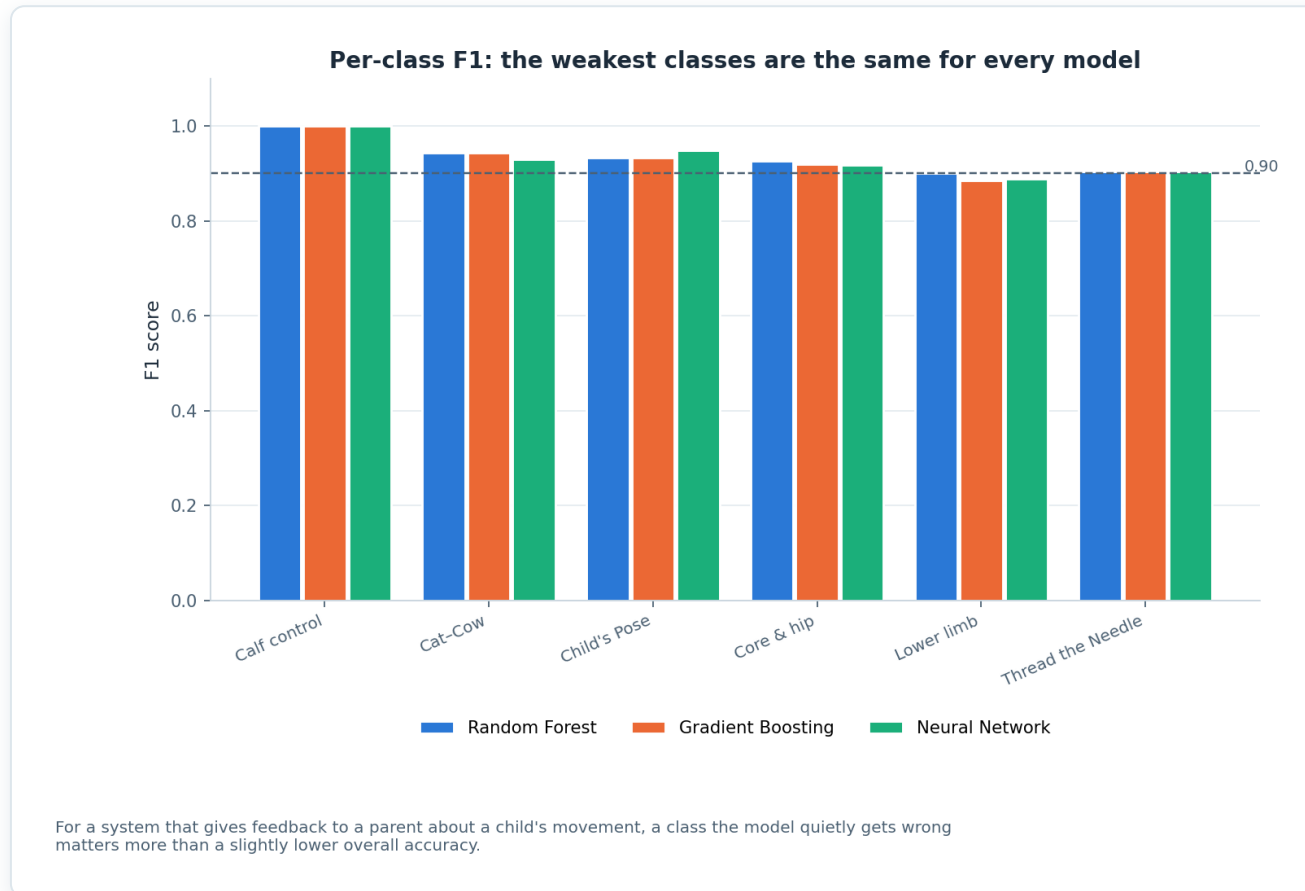


Figure 9 — per-class F1 across the three model families.

Consistent, not coincidental

- lower_limb and thread_the_needle are the weakest classes for every model.
- lower_limb sits at F1 approximately 0.89 to 0.90 depending on the model.
- thread_the_needle sits at F1 approximately 0.90 for all three.
- calf_control reaches 1.00 for all three, on a test support of 14 frames.

Precision, recall and F1 for every class

The selected model, scored once on 195 held-out test frames.

Class	Precision	Recall	F1	Support
calf_control	1.000	1.000	1.000	14
cat_cow	0.971	0.917	0.943	36
childs_pose	0.875	1.000	0.933	28
core_hip_stability	0.927	0.927	0.927	55
lower_limb	0.900	0.900	0.900	40
thread_the_needle	0.950	0.864	0.905	22

Macro F1 0.935, test accuracy 0.928. Measured on synthetic data.

FIGURE 8

The errors have a shape, and it is the expected one

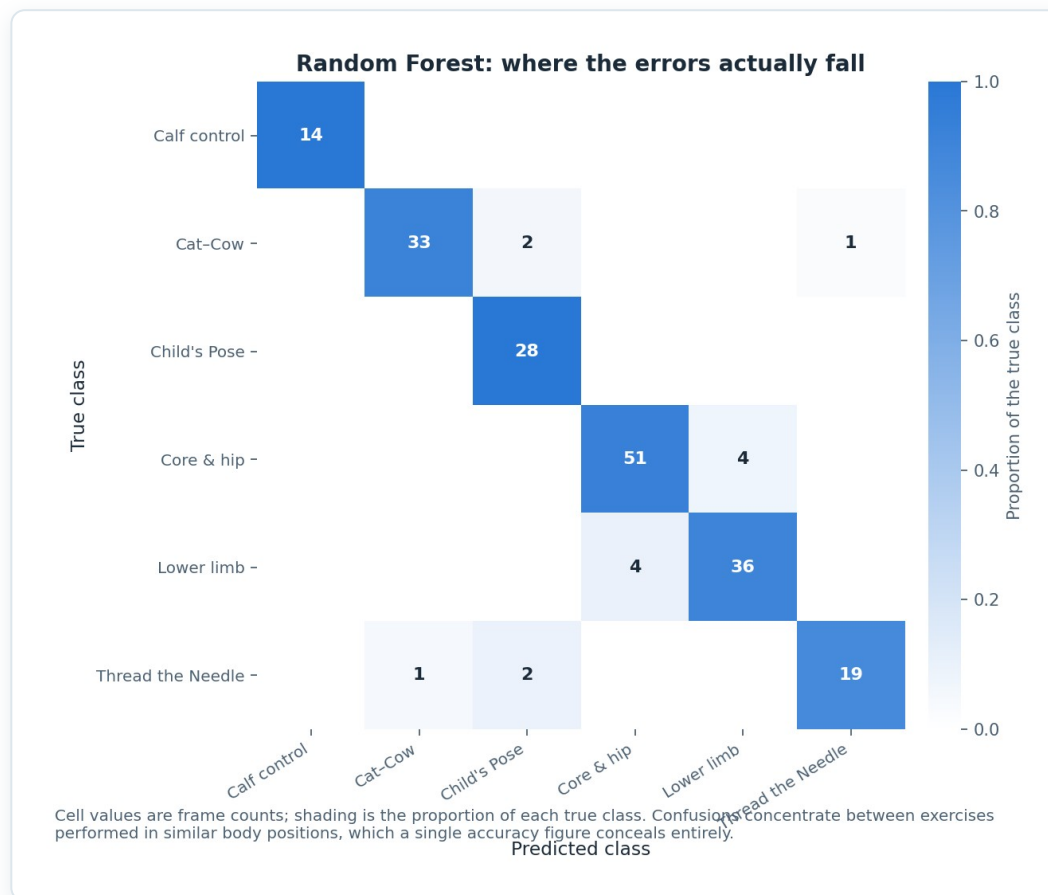


Figure 8 — Random Forest confusion matrix on the held-out test set.

What the matrix says

- Confusions concentrate between exercises performed in similar body positions.
- The quadruped-derived classes are confused with each other.
- The two lying-down classes are confused with each other.
- This structure was deliberately built into the generator, and it also appears in the PCA projection.

WHY THIS IS REASSURING

The errors are where a frame-level model should struggle: positions that look alike at a single instant. That is an argument for temporal features, not for a different classifier.

FIGURE 10

Normalised positions carry the model

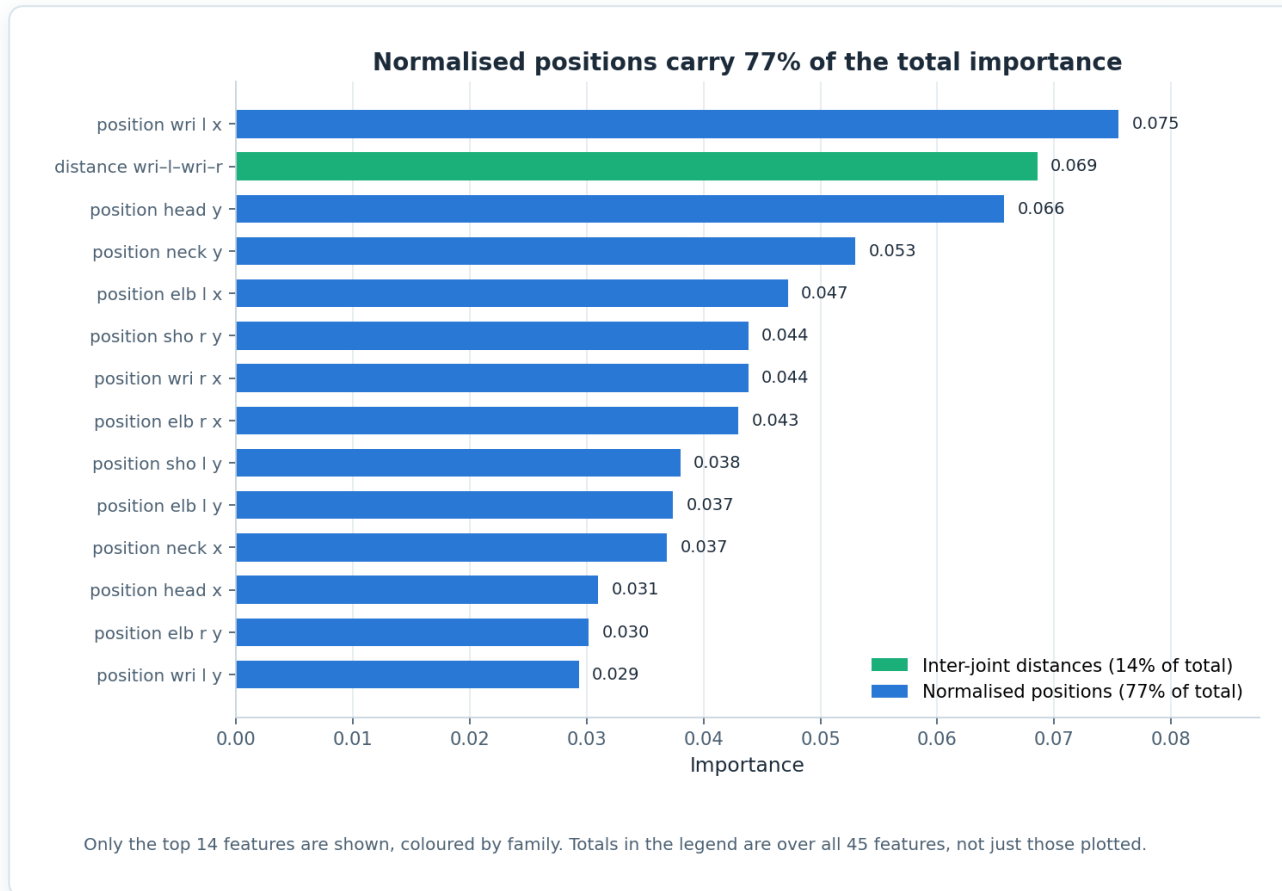


Figure 10 — top 14 features by Random Forest importance, coloured by family.

The share by family

- Normalised positions carry 77% of total importance.
- Inter-joint distances carry 14%.
- Angles contribute, but they do not dominate.
- The most useful features describe where the body is, once position and body size are normalised away.

A CORRECTION WE MAKE EXPLICITLY

It is tempting to say the model works because it measures joint angles. It does not. Positions at 77% and distances at 14% are the load-bearing families.

What this comparison cannot settle

- **The data is synthetic.**
Every score describes a seeded generator, not children performing physiotherapy.
- **The test set is small.**
195 frames in total, and calf_control has a support of 14 — its per-class figures are fragile.
- **Half a point of accuracy is not a real separation.**
The model choice rests on macro F1 and the generalisation gap, not on the accuracy column.
- **No temporal modelling.**
Frame-level features only, so the remaining confusions are the ones a movement-over-time model should resolve.
- **No clinical validation.**
No physiotherapist has reviewed the class definitions the model is trained to distinguish.

What would move the modelling forward

Ordered by expected effect on the numbers that currently limit the system.

1

Add temporal features over a sliding window

Velocity and phase across frames, then re-run this comparison unchanged. The confusion structure predicts exactly where the gain should appear.

2

Re-derive everything from a consented recorded dataset

Even 20 participants would replace every synthetic figure in this deck with a measured one.

3

Report calibrated confidence, not just a class

A system that gives feedback to a parent should be able to say when it is unsure.

4

Have a physiotherapist review the class definitions

The six classes come from published exercise text, and a clinical review would test whether they are the right six.

A comparison built to be believed, not to look good

Three families, one protocol

Identical data, identical splits, five-fold cross-validation on training only, and a test set scored exactly once.

Selected on the right criteria

Random Forest at 0.935 macro F1 with a +0.037 gap, chosen over two models that memorised their training data.

Reported without decoration

Per-class precision, recall and F1; the weakest classes named; positions at 77% of importance, not angles.

The number worth defending is not the highest one. It is the one produced by a protocol that could have embarrassed us.

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